

CCNA 200-301: Class Notes — Day 6

Topic: National Network Infrastructure & Telecom Architecture of Bangladesh

Bangladesh's telecommunications and internet infrastructure operates under a strictly regulated, multi-tiered framework established by the **Bangladesh Telecommunication Regulatory Commission (BTRC)**. This multi-tier architecture separates international bandwidth importation, nationwide transmission backhaul, local distribution, and inter-operator switching.

1. Regulatory Body: BTRC

- **BTRC (Bangladesh Telecommunication Regulatory Commission):**
 - **Role:** Statutory regulatory authority overseeing the entire telecommunication, broadcasting, and IP connectivity sector in Bangladesh.
 - **Responsibilities:**
 - Issuing licenses across all tiers (NTTN, IIG, ISP, IGW, ICX, MNO).
 - Radio spectrum allocation for mobile operators (3G, 4G, 5G) and Wi-Fi/satellite links.
 - Tariff regulation and monitoring Quality of Service (QoS).
 - Enforcement of national cyber safety and ILDTS (International Long Distance Telecommunications Services) policies.

2. International Bandwidth Ingress: Submarine Cables & ITC

Bangladesh receives international Internet bandwidth through two primary physical pathways: **Submarine Cables** (sea-based) and **International Terrestrial Cables** (land-based cross-border links).

A. BSCCL & Submarine Cables (SEA-ME-WE)

- **BSCCL (Bangladesh Submarine Cables PLC):** State-owned Core

Telecommunications Service Provider operating Bangladesh's submarine cable landing stations.

- **Submarine Cable Systems:**

1. **SMW-4 (SEA-ME-WE 4):** Connected in 2006 with a landing station at **Cox's Bazar**. Provides primary legacy bandwidth to South Asia, Middle East, and Europe.
2. **SMW-5 (SEA-ME-WE 5):** Connected in 2017 with a landing station at **Kuakata (Patuakhali)**. Provides ultra-low latency, high-capacity connectivity.
3. **SMW-6 (SEA-ME-WE 6):** Third-generation high-capacity submarine cable system landing at **Cox's Bazar**, drastically expanding national terabit-scale capacity.

B. ITC (International Terrestrial Cable)

- **Role:** Terrestrial optical fiber cables crossing land borders (primarily connecting Bangladesh with India to reach global submarine cable landing stations in Mumbai and Chennai).
- **Key Function:** Acts as an active redundant failover for submarine cable cuts and expands total wholesale bandwidth capacity.
- **Licensed Operators:** Private infrastructure entities such as Summit Communications, Fiber@Home, Mango Teleservices, BD Link, and NovoCom.

3. Gateway Layer: IIG & IGW

To maintain security and structured routing, bandwidth imported from international links must pass through licensed gateways.

Gateway Entity	Acronym	Primary Function & Layer	Key Examples
International Internet	IIG	Processes, aggregates,	Summit Communicatio

Gateway		and routes IP internet data traffic between international sources (BSCCL/ITC) and local ISPs/Mobile Operators.	ns, Fiber@Home, BTCL, Mango Teleservices, Novocom.
International Gateway	IGW	Routes incoming and outgoing international voice calls between foreign carriers and national interconnection exchanges (ICX).	Unique Infoway, Mir Telecom, Roots Communication, BTCL.

4. Transmission Backhaul: NTTN Layer

- **NTTN (Nationwide Telecommunication Transmission Network):**
 - **Role:** Licensed operators authorized to build, maintain, and lease nationwide dark fiber, DWDM, and IP-MPLS backhaul transmission lines.
 - **Why NTTN exists:** Regulatory policies prohibit individual ISPs and Mobile Operators from laying their own long-haul underground optical fiber across districts, preventing duplicate trenching and uncoordinated infrastructure expansion.
 - **Major NTTN Operators:**

- Summit Communications Ltd.
- Fiber@Home Ltd.
- BTCL (Bangladesh Telecommunications Company Limited)
- PGCB (Power Grid Company of Bangladesh - leases utility fiber)

5. Voice Routing: ICX Layer

- **ICX (Interconnection Exchange):**
 - **Role:** Centralized voice routing and switching switches for inter-operator calls within Bangladesh.
 - **Functionality:** When a user calls from Grameenphone to Robi, or from a landline to a mobile network, the voice traffic passes through an ICX to handle call setups, inter-operator settlement billing, and routing to IGWs for foreign calls.

6. Access Network Services: ISPs & Mobile Operators

Access Network Service (ANS) providers deliver the final connection ("last-mile delivery") to residential, corporate, and commercial users.

A. ISPs (Internet Service Providers)

- **Categories:**
 - **Nationwide ISPs:** Licensed to operate across all 64 districts in Bangladesh (e.g., Amber IT, Link3, Carnival Internet, BDCOM, Dotlines).
 - **Zonal / Divisional / Category A, B, C ISPs:** Regional broadband operators delivering Fiber-to-the-Home (FTTH) and local ethernet connections.

B. Mobile Network Operators (MNOs)

- **Operators:** Grameenphone (GP), Robi Axiata, Banglalink, and Teletalk.
- **Services:** Wireless data services (3G, 4G LTE, 5G trials), Mobile Financial Services (MFS) connectivity, and cellular voice over radio access networks (RAN).

7. Local Peering & Exchange: IXP, NIX, and BDIX

Without local internet exchanges, two users in Dhaka communicating with each other over different ISPs would have their traffic routed through international undersea hops (e.g., Singapore or India), creating massive latency and consuming international bandwidth.

[Local ISP A User] ———▶ [BDIX / NIX Exchange Switch] ———▶ [Local ISP B User]

(Sub-5ms Latency)

- **IXP (Internet Exchange Point) / NIX (National Internet Exchange):** Physical Layer 2/3 network switching fabrics that allow local ISPs, MNOs, CDNs, and content providers to exchange local internet traffic directly without sending it over expensive international links.
- **BDIX (Bangladesh Development Computing and Internet Exchange):**
 - **Significance:** The pioneer and premier National IXP in Bangladesh.
 - **Key Benefits:**
 - Reduces latency for domestic traffic down to < 5ms.
 - Keeps domestic server traffic (FTP, local news, local streaming, banking portals, government sites) local.
 - Saves millions of dollars in foreign currency by reducing international bandwidth consumption.

Summary Matrix of Bangladesh Telecom Entities

Acronym	Full Form	Industry Hierarchy Tier
BTRC	Bangladesh Telecommunication Regulatory Commission	Policy & Regulatory Body

BSCCL	Bangladesh Submarine Cables PLC	International Submarine Landing Authority
ITC	International Terrestrial Cable	Cross-Border Terrestrial Bandwidth Importer
IGW	International Gateway	International Voice Call Routing Gateway
IIG	International Internet Gateway	Wholesale IP Bandwidth Aggregator
NTTN	Nationwide Telecommunication Transmission Network	Physical Fiber Optic Backbone Provider
ICX	Interconnection Exchange	Inter-operator Local Voice Switching Hub
ISP	Internet Service Provider	Broadband Access Provider (FTTH / Enterprise)
IXP / NIX / BDIX	National Internet Exchange / Bangladesh IX	Domestic Traffic Peering & Routing Fabric